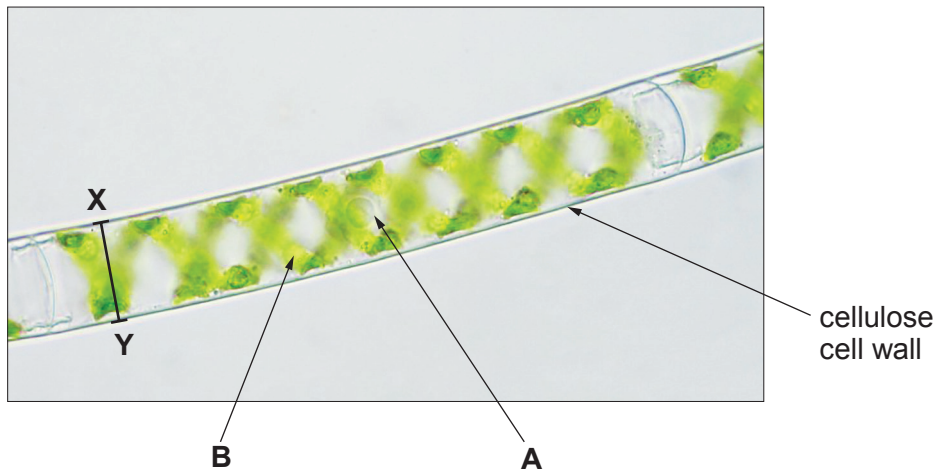


3. The photomicrograph below is of *Spirogyra*; an autotrophic organism that inhabits fresh water ponds and ditches.



(a) Identify the organelles labelled **A** and **B**. [2]

A:

B:

(b) The actual width of the cell between points **X** – **Y** was 32.3 μm . Calculate the magnification that was used to take the photomicrograph. [2]

magnification =

(c) During the day the concentration of solutes in the pond water changes due to evaporation. However, the length of the *spirogyra* cells remain almost constant. What can you conclude about how the structure of the cell wall enables *Spirogyra* to survive in different solute concentrations? [4]

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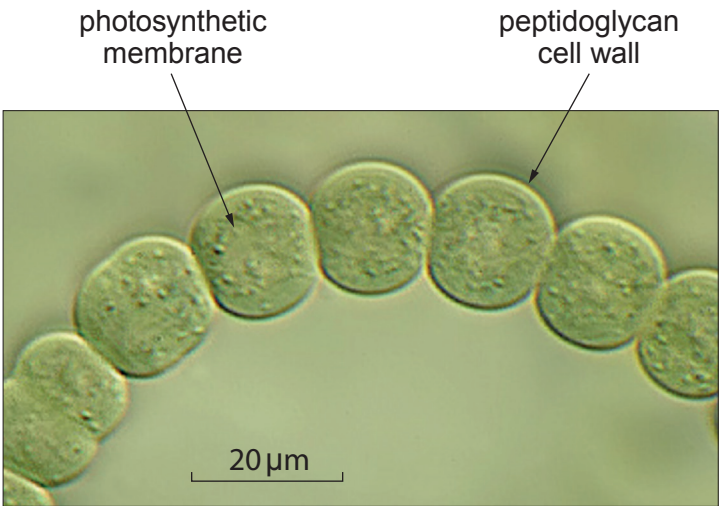
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(d) Below is a photomicrograph of the bacteria *Nostoc*, which also inhabits fresh water ponds and ditches. They were once thought to belong to the same group of organisms as *Spirogyra* and thought to have the same cellular structure. Evidence from electron microscopy has now grouped these two separately.

Conclude what cell types are present in *Spirogyra* and *Nostoc*. Identify **two** differences and **one** similarity (not labelled in the photomicrographs) between these two species that would be revealed by electron microscopy. [3]



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